





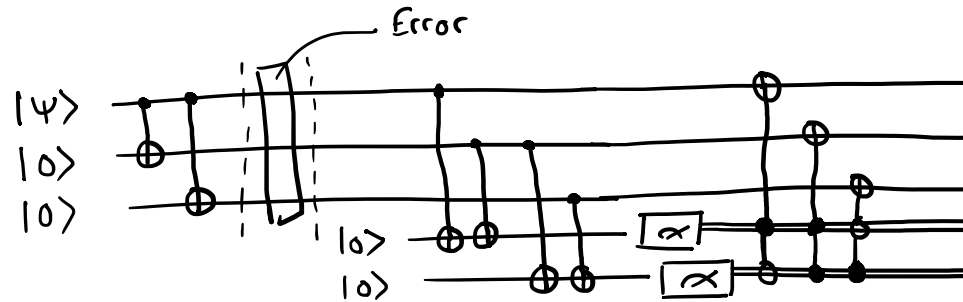
Quantum Error Mitigation(II)

Phase Flip Error Mitigation

Pinakin Padalia

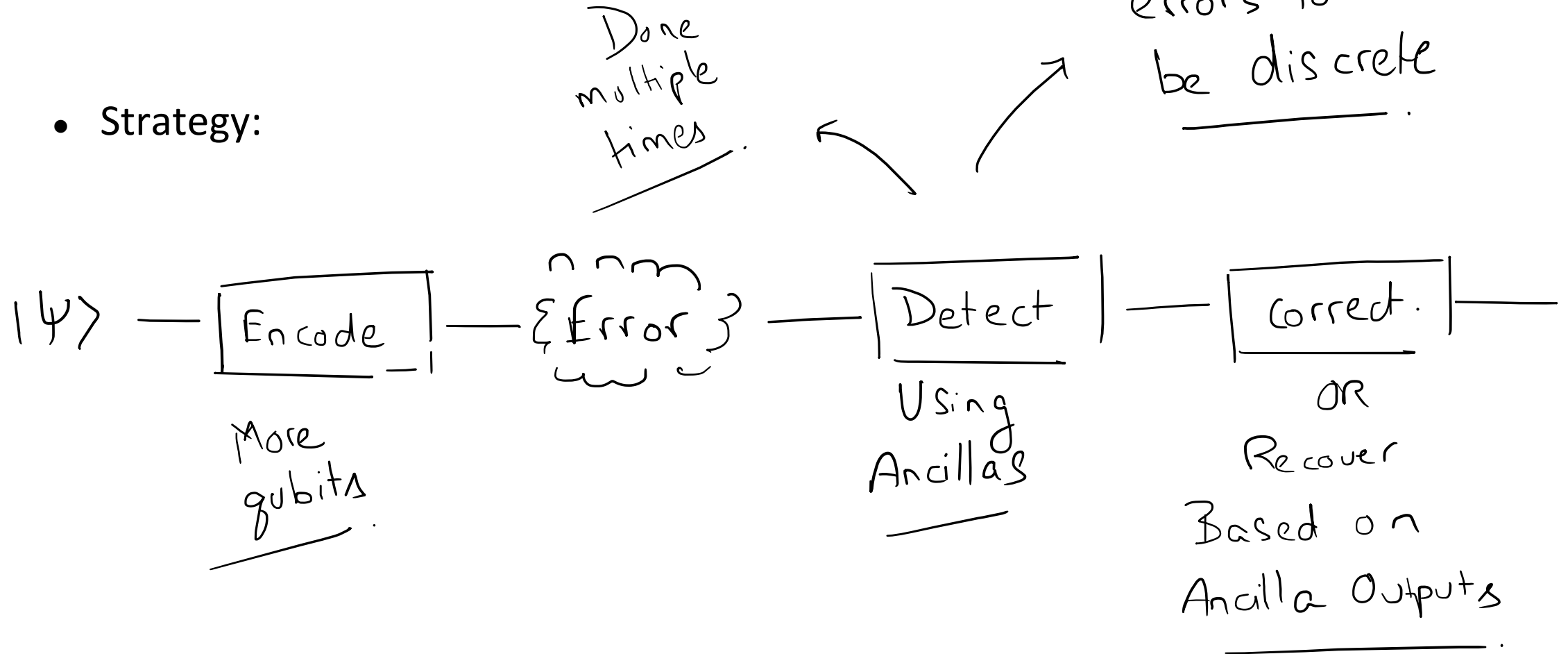
What are we covering in this chapter?

- Bit Flip Error Mitigation
- **Phase Flip Error Mitigation:**
 - Strategy for error mitigation: Recap
 - Error Mitigation steps: 3 – qubit phase flip code
 - Limitations
- Shor code (9 qubit code) and fault tolerance



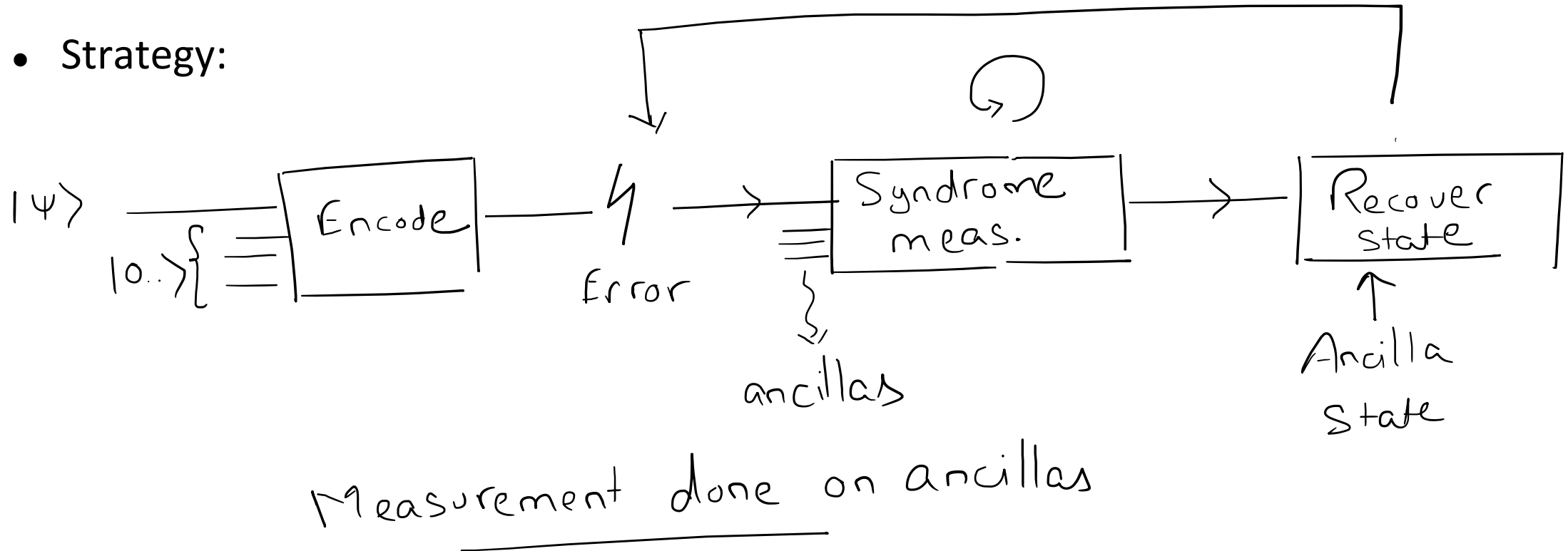
Strategy for error mitigation

- Strategy:



Strategy for error mitigation: Recap

- Strategy:



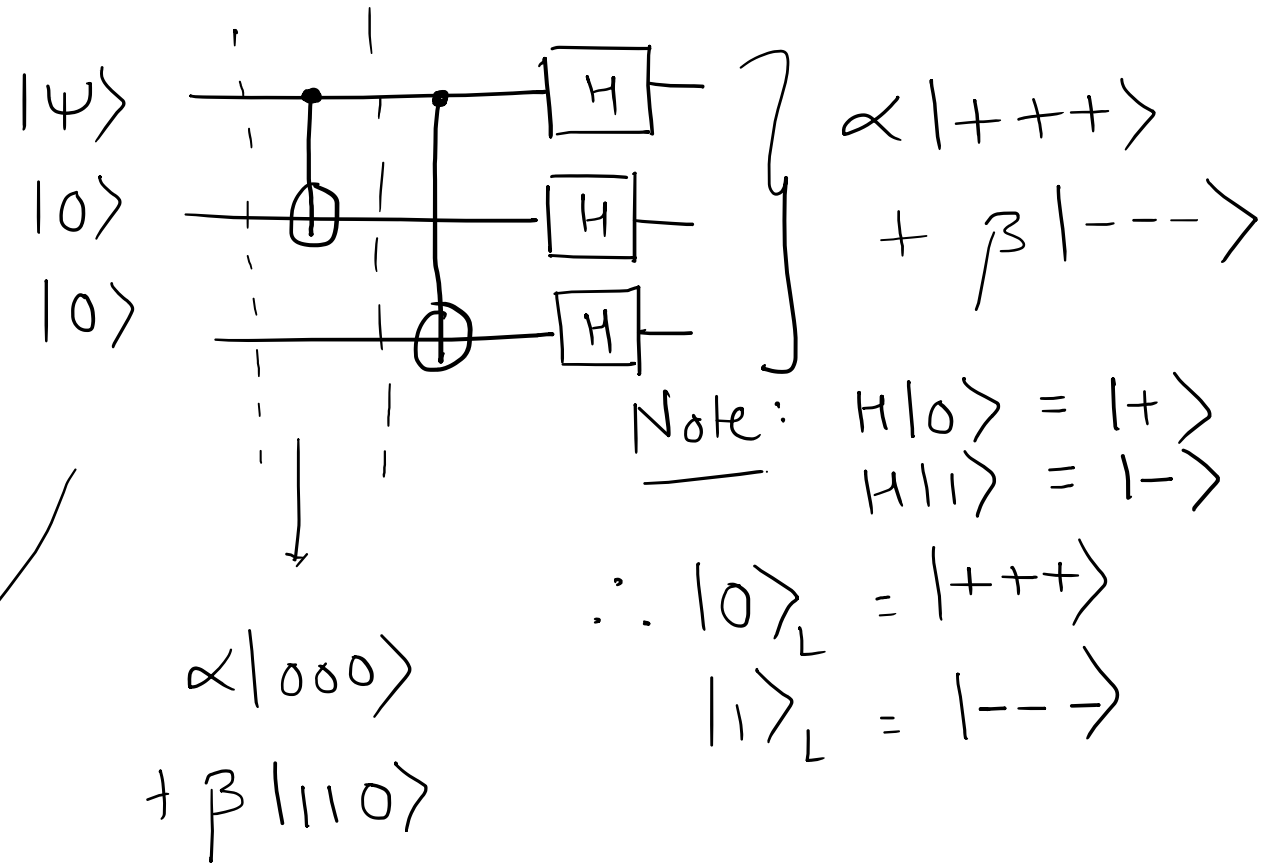
- Phase flip error: Pauli Z error
- Mitigation: 3 – qubit phase flip code
 - Step 1: Encode

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

Phase flip in
 $|0\rangle, |1\rangle$ basis
 $\equiv$

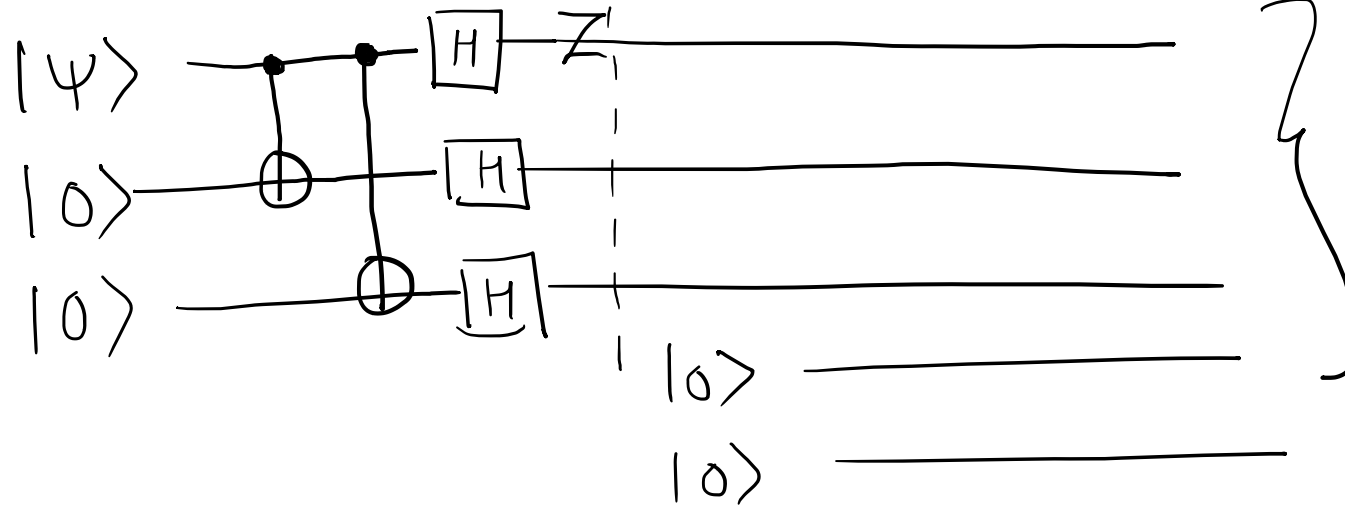
Bit flip in
 $|+\rangle, |-\rangle$ basis

$$\alpha|000\rangle + \beta|100\rangle$$



Phase Flip Error Mitigation

- Phase flip error: Pauli Z error
- Mitigation:
 - Step 2: Add ancillas



$\alpha |-\bar{+}\bar{+}\rangle + \beta |+\bar{-}\bar{-}\rangle$
 equal? — Yes — No
 Bit flip

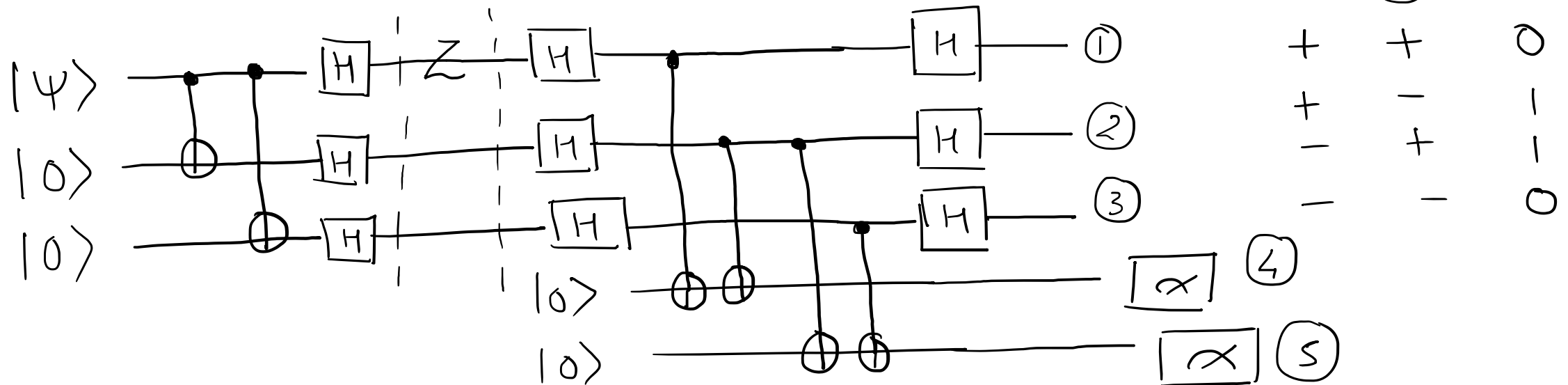
Bit flip!
 in $|+\rangle, |-\rangle$
 Basis

$$\alpha |-\bar{+}\bar{+}\underline{00}\rangle + \beta |+\bar{-}\bar{-}\underline{00}\rangle$$

Phase Flip Error Mitigation

- Phase flip error: Pauli Z error
- Mitigation:
 - Step 3: Syndrome Measurement

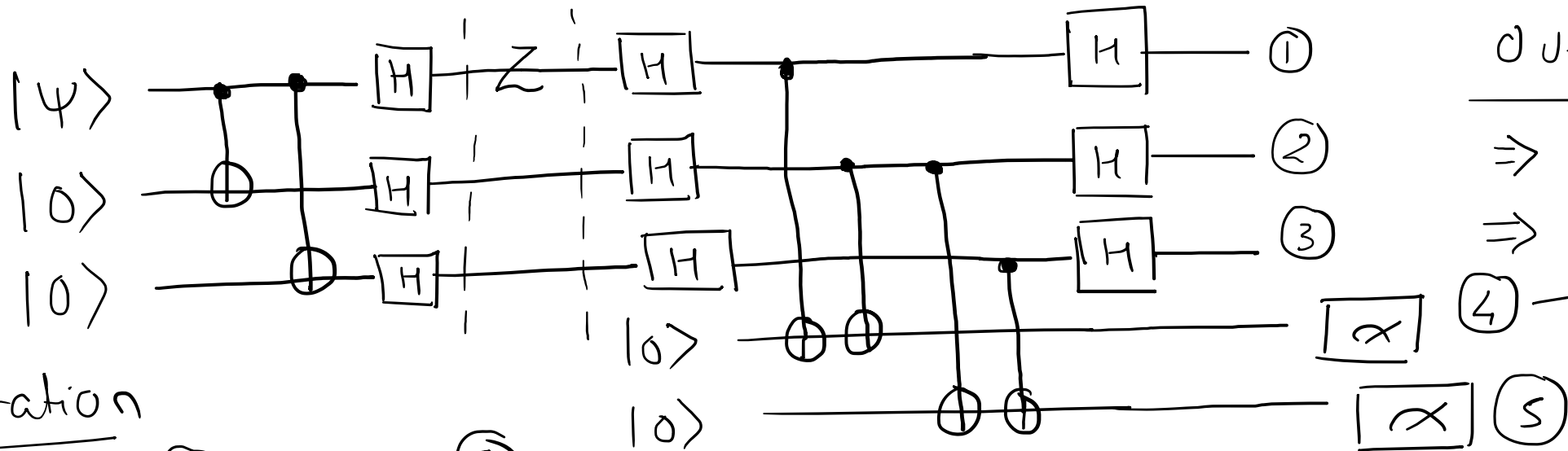
$\alpha |+-+\rangle + \beta |+--\rangle$
 equal? — Yes — No
 Bit flip
 Bit flip! in $|+\rangle, |-\rangle$ Basis



Phase Flip Error Mitigation

- Phase flip error: Pauli Z error
- Mitigation:
 - Step 3: Syndrome Measurement

$\alpha|+-+\rangle + \beta|+--\rangle$
 equal? — Yes — No
 Bit flip
 Bit flip! in $|+\rangle, |-\rangle$ Basis



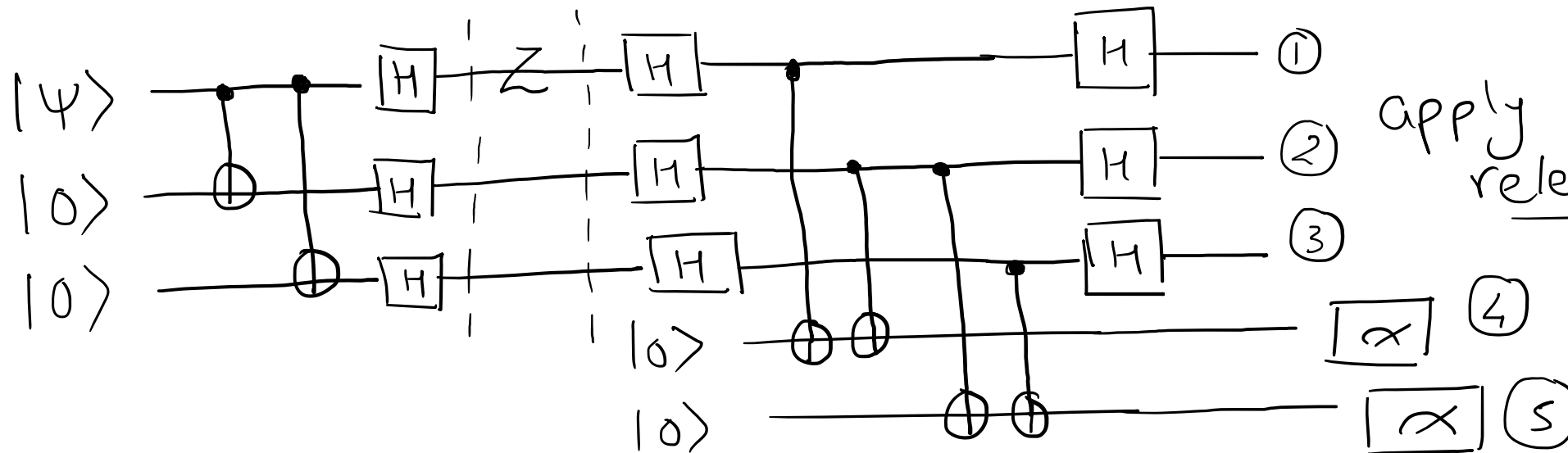
Measured
 output
 $\Rightarrow 1 = \text{Bit flip}$
 $\Rightarrow 0 = \text{No error}$

Notation

$$x_1 x_2 = 4; x_2 x_3 = 5$$

Phase Flip Error Mitigation

- Phase flip error: Pauli Z error
- Mitigation:
 - Step 4: Recovery



(4)	(5)	
0	0	→ No error
1	0	→ error in (1)
0	1	→ error in (3)
1	1	→ error in (2)

apply Z gate to relevant qubit.

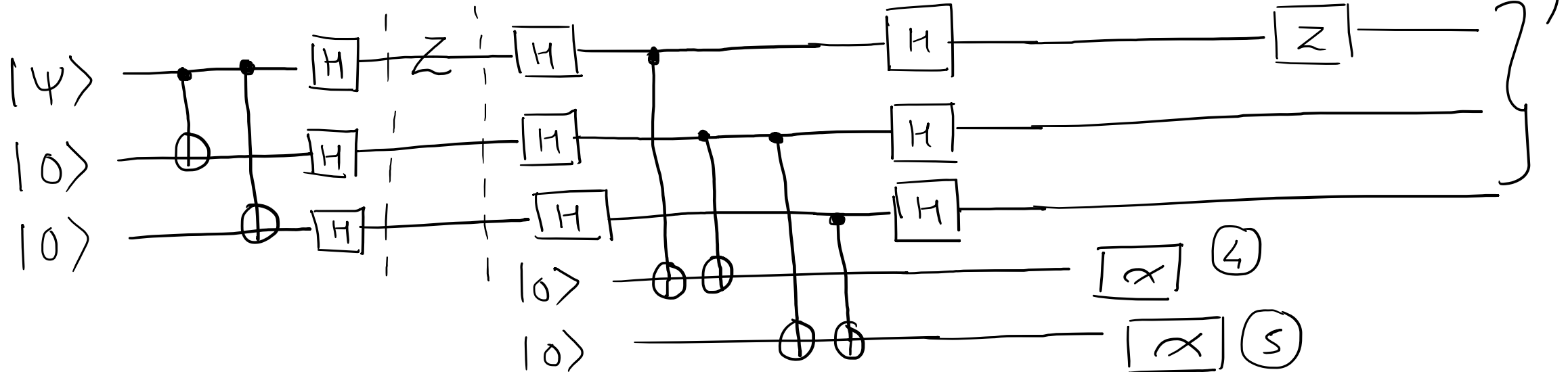
Phase Flip Error Mitigation

- Phase flip error: Pauli Z error
- Mitigation:
 - Step 4: Recovery

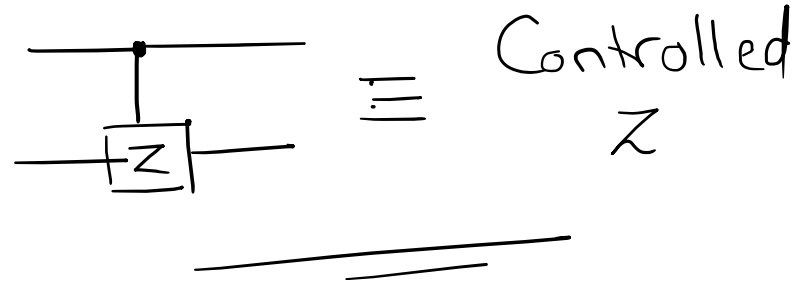
④ ⑤
 1 0

Apply z gate to qubit ①

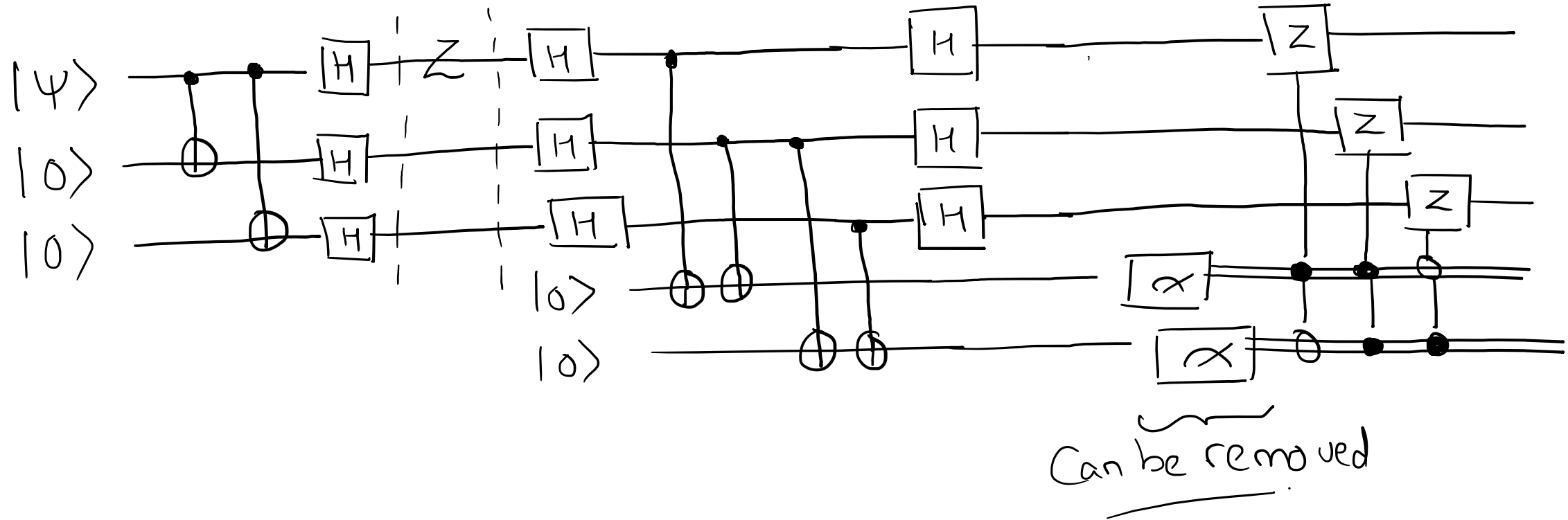
$$\propto |+++ \rangle + \beta |--- \rangle$$



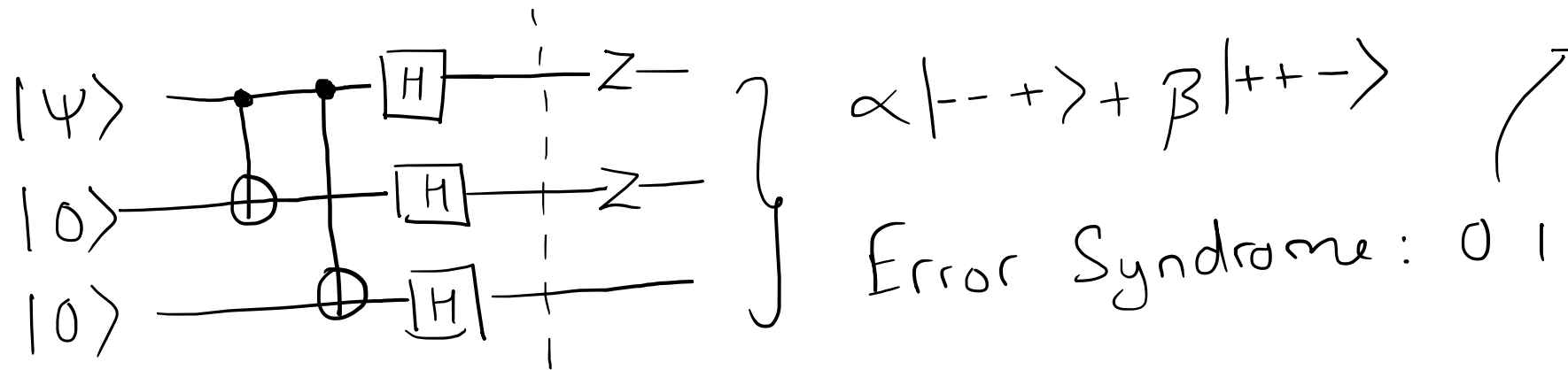
Phase Flip Error Mitigation



- Phase flip error: Pauli Z error
- Mitigation: generalized picture



- Phase flip error: Pauli Z error
- Mitigation: Limitations
 - Requires more qubits – a scarce resource!
 - Requires repeated operations – problem with short coherence times!
 - Not effective for more than one error



correct qubit 3!



Thank you for your attention!

- Michael A. Nielsen and Isaac L. Chuang. 2011. Quantum Computation and Quantum Information: 10th Anniversary Edition (10th. ed.). Cambridge University Press, USA.
- M. Wilde, "From Classical to Quantum Shannon Theory", Arxiv.org, 2019. [Online]. Available: <https://arxiv.org/pdf/1106.1445.pdf>.
- J. Preskill, "Lecture Notes for Physics 229: Quantum Information and Computation", Lorentz.leidenuniv.nl, 1998. [Online]. Available: https://www.lorentz.leidenuniv.nl/quantumcomputers/literature/preskill_1_to_6.pdf.